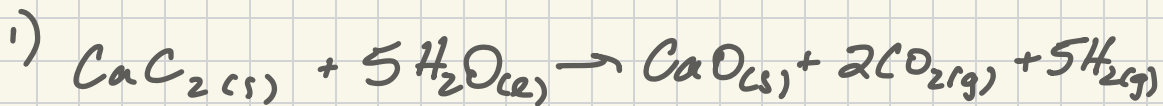


# Homework 10



$$\Delta H^\circ_{\text{rxn}} = +69.36 \text{ kJ/mol} \checkmark$$

- a) B/c  $\Delta H^\circ_{\text{rxn}} > 0$ , the rxn is endothermic, and you would have to heat the reactor to keep temperature constant. If the reactor was kept adiabatic, meaning no heat could enter or exit the system, the temp. of the system would decrease. The reaction requiring heat to proceed implies the reactants are more stable than the products being formed. ✓

b)  $\Delta U_r^\circ = \Delta H^\circ - \Delta n_g RT$  20/20  $\Delta n_g = 7$

$$\Delta U_r^\circ = 69.36 - 7(0.008314)(298)$$

✓  $\Delta U_r^\circ = 52.22 \text{ kJ/mol}$  ← energy needed to be supplied for rxn to proceed



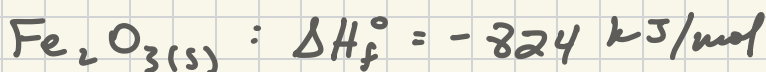
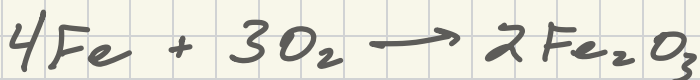
$$Q = n \Delta U_r^\circ = 2.34(52.22) = 121.7 \text{ kJ into sys.}$$

2) Murphy 6.34

$$q = m_{H_2O} C_p \Delta T = (650)(4.184)(12.6) \\ = 34.3 \text{ kJ}$$

$$\Delta U_c = 34.3 / 2.7 = \boxed{-12.7 \text{ kJ/g}} \quad \checkmark \quad 20/20 \\ \text{or } \boxed{-1.96 \times 10^3 \text{ kJ/mol}}$$

3) Murphy 6.34



$$2 \text{ mol} \times -824 = -1648 \text{ kJ}$$

$$-1648 \text{ kJ} / 4 \text{ mol} = 412 \text{ kJ/mol Fe}$$

$$Q = \frac{100 \text{ kJ}}{\text{g}} \times 6 \text{ g} = 600 \text{ kJ} \quad 20/20$$

$$600 \text{ kJ} \times \frac{1 \text{ mol Fe}}{412 \text{ kJ}} = 1.456 \text{ mol Fe}$$

$$1.456 \text{ mol} \times \frac{55.85 \text{ g}}{1 \text{ mol}} = \boxed{81.4 \text{ g Fe}} \quad \checkmark$$

$$1.456 \text{ mol Fe} \times \frac{3 \text{ mol O}_2}{4 \text{ mol Fe}} \times \frac{32 \text{ g O}_2}{1 \text{ mol O}_2} = 34.94 \text{ g O}_2$$

$$1.456 \text{ mol Fe} \times \frac{1 \text{ mol Fe}_2\text{O}_3}{2 \text{ mol Fe}} \times \frac{159.69 \text{ g}}{1 \text{ mol Fe}_2\text{O}_3} = 116.3 \text{ g Fe}_2\text{O}_3$$

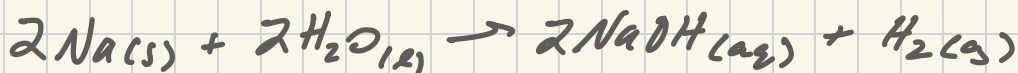
$$\Delta m = 116.3 - 81.4 = 34.9 \text{ g} = \Delta m$$

$$\frac{34.9}{81.4} = 43\% \text{ increase} \quad \checkmark$$

4)

-20

5) Murphy 5.37



$$\text{NaOH}(aq) : \Delta H_f^\circ = -469.6 \text{ kJ/mol}$$

$$\text{H}_2\text{O}_{(l)} : \Delta H_f^\circ = -285.8 \text{ kJ/mol}$$

$$\begin{aligned}\Delta H_{\text{rxn}}^\circ &= 2(-469.6) - 2(-285.8) / 2 \\ &= -183.8 \text{ kJ/mol Na}\end{aligned}$$

$$100\text{g Na} \rightarrow 4.35 \text{ mol Na}$$

$$1\text{g H}_2\text{O} \rightarrow 0.056 \text{ mol H}_2\text{O}$$

$$Q = 183.8(0.056) = 10.2 \text{ kJ heat released} \checkmark$$

The rxn is very exothermic & can release 10.2 kJ of heat, which can boil the water & ignite the flammable  $\text{H}_2$  gas.

20/20